

Mutual Fund Analytics Dashboard & Performance Evaluation

Capstone Project - Mutual Fund Analytics

Presented by

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Blue Stock Fintech Internship Program

Capstone Project I

June 2026

Project Overview

Objective:

- ▶ Analyze mutual fund performance using historical NAV data.
- ▶ Evaluate risk-adjusted returns using advanced financial metrics.
- ▶ Compare fund performance against benchmark indices.

Tools & Technologies:

- ▶ Python
- ▶ Pandas
- ▶ NumPy
- ▶ Matplotlib
- ▶ Github

Project Coverage:

- ▶ 40 Mutual Fund Schemes
- ▶ Historical Data (2022–2025)
- ▶ NIFTY 50 & NIFTY 100 Benchmarks
- ▶ Performance & Risk Analytics

Data Sources & Project Architecture

Data Sources:

Mutual Fund Data:

- ▶ Fund Master Data
- ▶ NAV History Data
- ▶ AUM Data
- ▶ SIP Inflows Data
- ▶ Investor Transaction Data

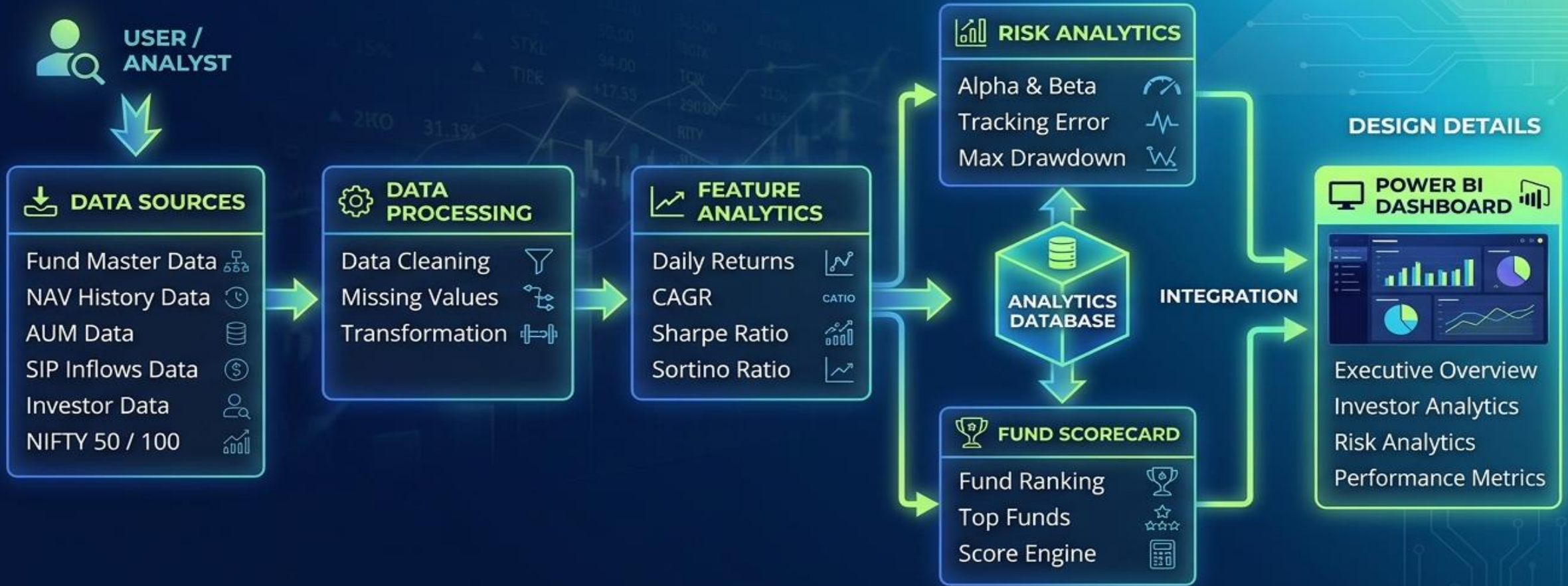
Benchmark Data:

- ▶ 40 Mutual Fund Schemes
- ▶ Historical Data (2022–2025)
- ▶ NIFTY 50 & NIFTY 100 Benchmarks
- ▶ Performance & Risk Analytics

Project Architecture

MUTUAL FUND ANALYTICS ARCHITECTURE

High-resolution professional PowerPoint



Data Cleaning & Exploratory Data Analysis (EDA)

Data Cleaning Activities:

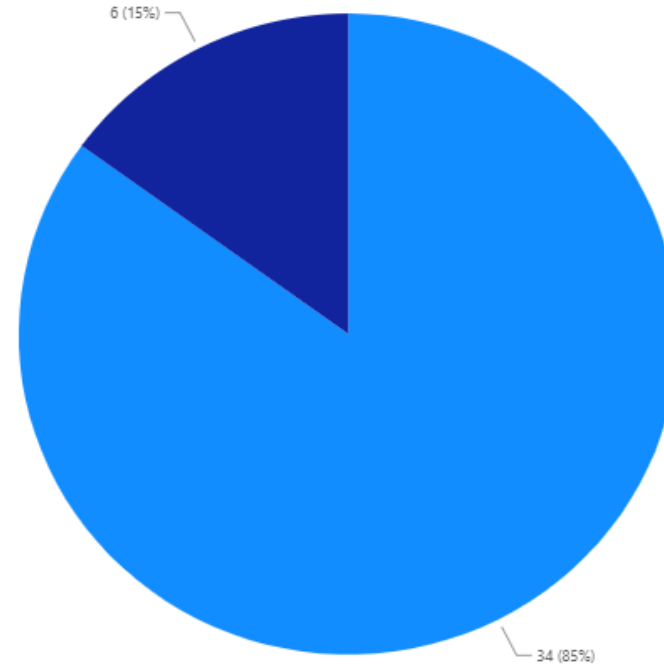
- ▶ Removed duplicate records.
- ▶ Handled missing and inconsistent values.
- ▶ Standardized date formats and column names.
- ▶ Validated data quality across datasets.

Exploratory Data Analysis:

- ▶ Fund category distribution analysis.
- ▶ SIP inflow trend analysis.
- ▶ AUM growth analysis.
- ▶ Investor demographic analysis.

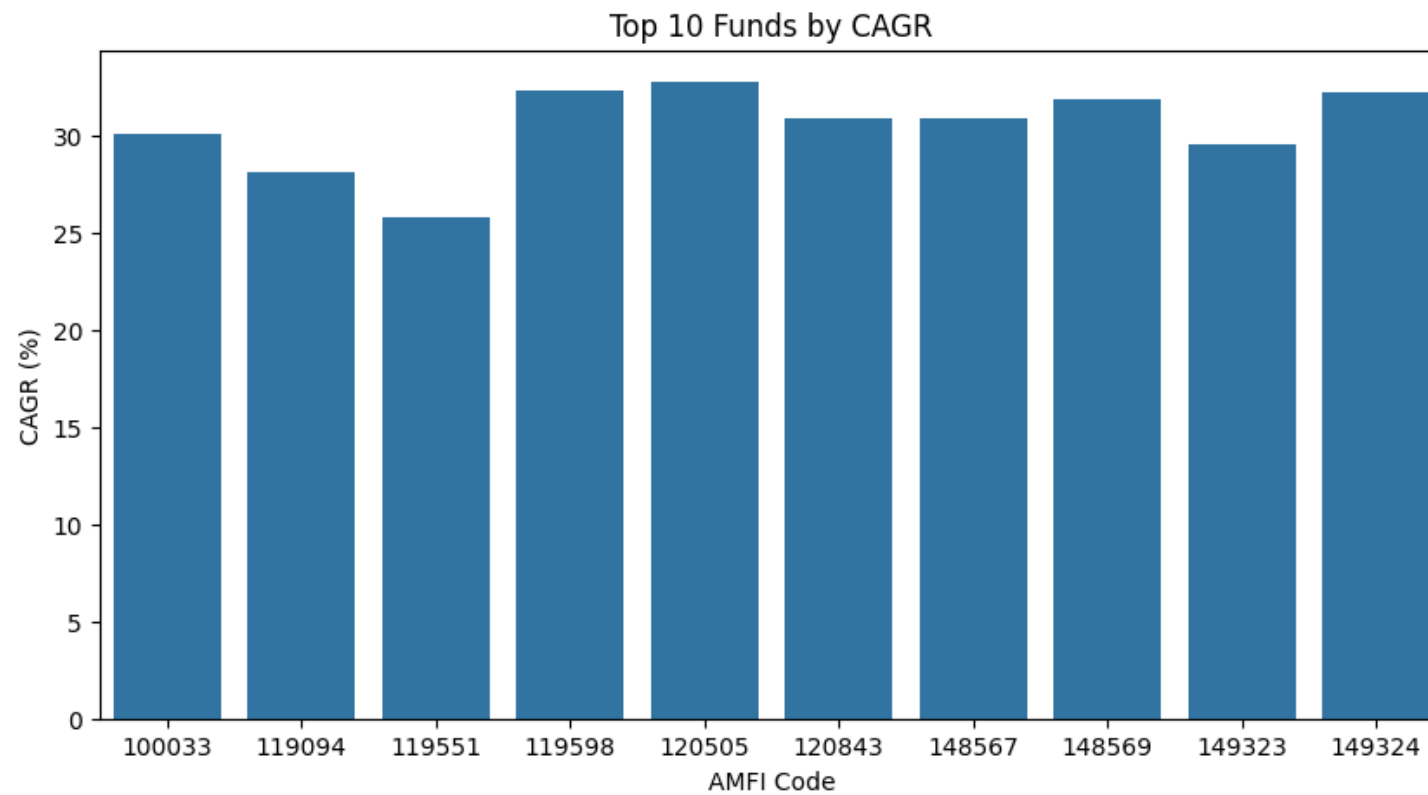
Fund Score Distribution Pie Chart

Fund Category Distribution



category
● Equity
● Debt

CAGR Results



Performance Analytics Metrics

Metrics Calculated:

- ▶ Measured the excess return generated by a mutual fund compared to its benchmark index.
- ▶ Helped identify funds that consistently outperformed market expectations.

Methodology:

- ▶ Calculated daily returns from NAV history.
- ▶ Measured long-term growth using CAGR.
- ▶ Evaluated risk-adjusted returns using Sharpe Ratio.
- ▶ Assessed downside risk using Sortino Ratio.

Key Outcomes:

- ▶ Identified top-performing mutual funds.
- ▶ Compared fund performance across different categories.
- ▶ Generated performance rankings based on calculated metrics.

Fund Performance Ranking

Sum of fund_score by scheme_name



Risk Analytics & Benchmark Evaluation

Risk Metrics Calculated:

Alpha:

- ▶ Measured the excess return generated by a mutual fund compared to its benchmark index.
- ▶ Helped identify funds that consistently outperformed market expectations.

Beta:

- ▶ Evaluated the sensitivity of a fund's returns relative to market movements.
- ▶ Used to assess the systematic risk associated with each fund.

Tracking Error:

- ▶ Measured the deviation between fund returns and benchmark returns.
- ▶ Helped analyze the consistency of benchmark tracking performance.

Maximum Drawdown:

- ▶ Calculated the maximum observed loss from a peak to a trough during the investment period.
- ▶ Provided insights into downside risk and capital preservation capability.

Key Findings:

Benchmark Comparison:

- ▶ Alpha and Beta metrics enabled effective comparison of fund performance against benchmark indices.

Tracking Efficiency:

- ▶ Tracking Error analysis identified funds with better benchmark replication and consistency.

Downside Risk Assessment:

- ▶ Maximum Drawdown highlighted the potential losses investors could experience during adverse market conditions.

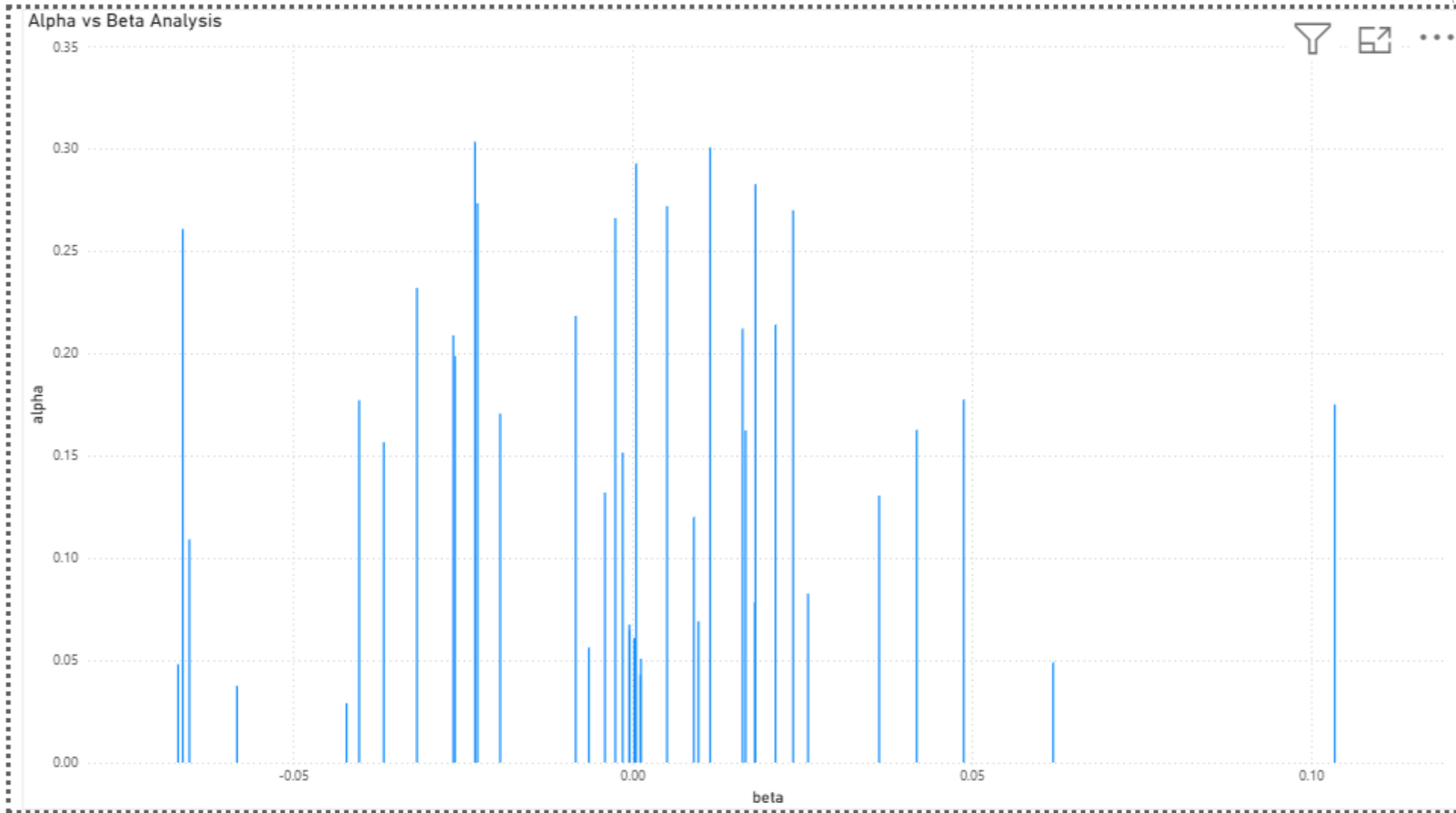
Risk-Adjusted Performance:

- ▶ Combined risk metrics provided a comprehensive understanding of fund performance beyond returns alone.

Investment Insights:

- ▶ Risk analytics helped identify funds with balanced return potential and manageable risk exposure.

Tracking Error Analysis



Fund Scorecard & Top Performing Funds

Fund Scoring Methodology:

A composite fund score was generated using:

- ▶ CAGR
- ▶ Sharpe Ratio
- ▶ Sortino Ratio
- ▶ Alpha
- ▶ Tracking Error
- ▶ Maximum Drawdown

These metrics were combined to rank mutual funds based on both return potential and risk management.

Top Performing Funds:

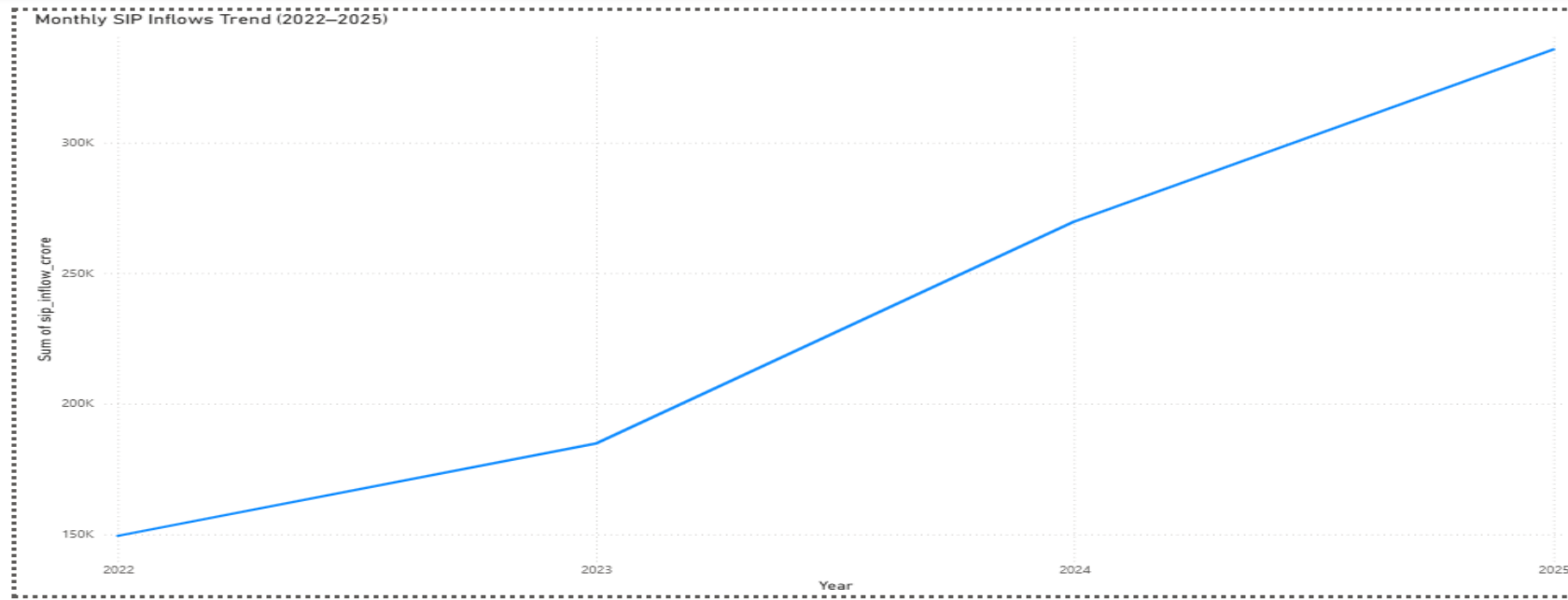
- ▶ ICICI Pru Midcap Fund
- ▶ Kotak Flexicap Fund
- ▶ HDFC Mid-Cap Opportunities Fund
- ▶ SBI Small Cap Fund

Key Findings:

- ▶ ICICI Pru Midcap Fund achieved the highest overall fund score.
- ▶ Mid-cap and small-cap funds demonstrated strong performance potential.
- ▶ Risk-adjusted metrics helped identify consistently performing schemes.
- ▶ Fund scorecard simplified comparison across multiple mutual fund schemes.

Rank	Fund
1	ICICI Pru Midcap Fund
2	Kotak Flexicap Fund
3	HDFC Mid-Cap Opportunities Fund
4	SBI Small Cap Fund
5	Mirae Asset Large Cap Fund

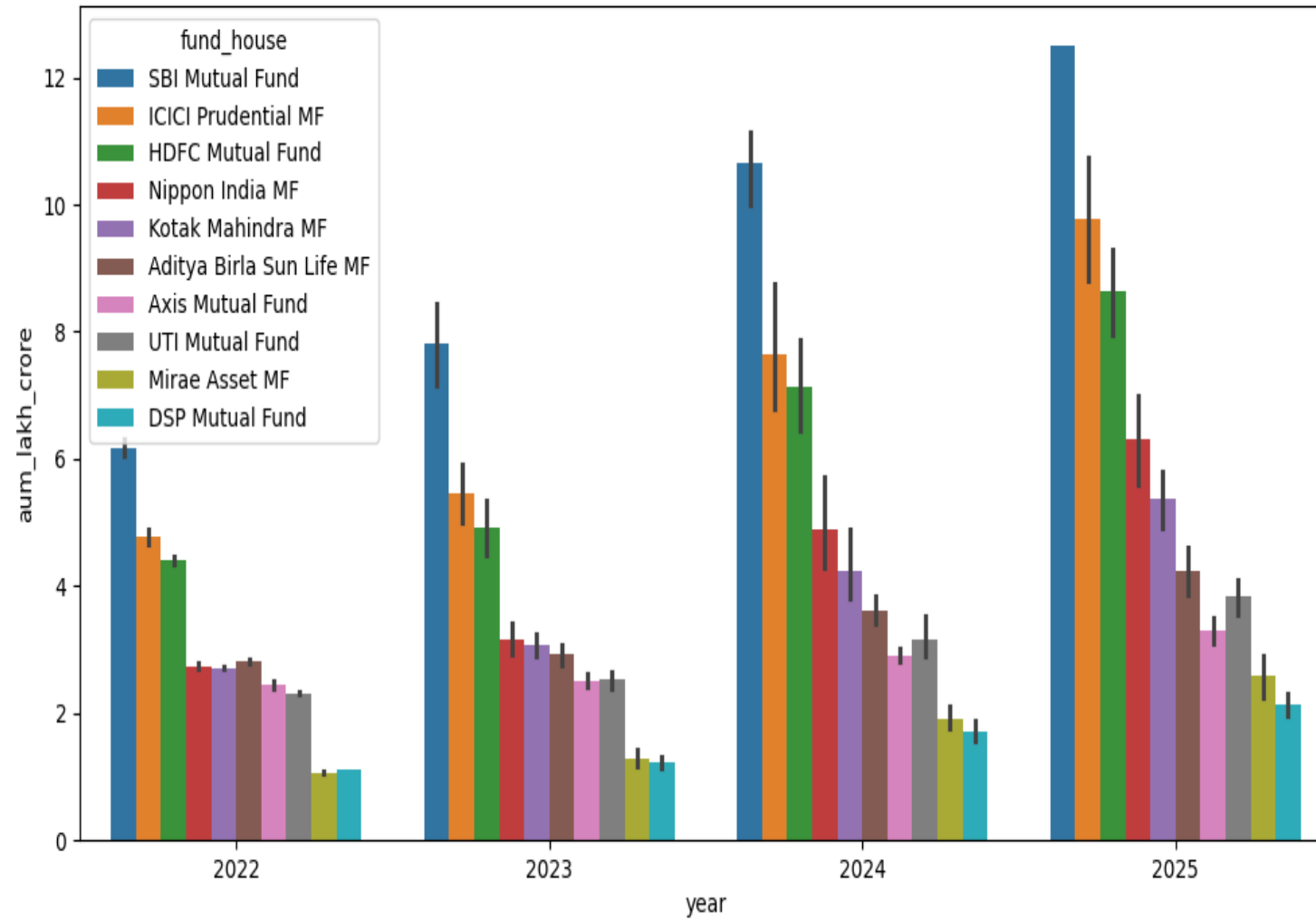
Executive Overview Dashboard



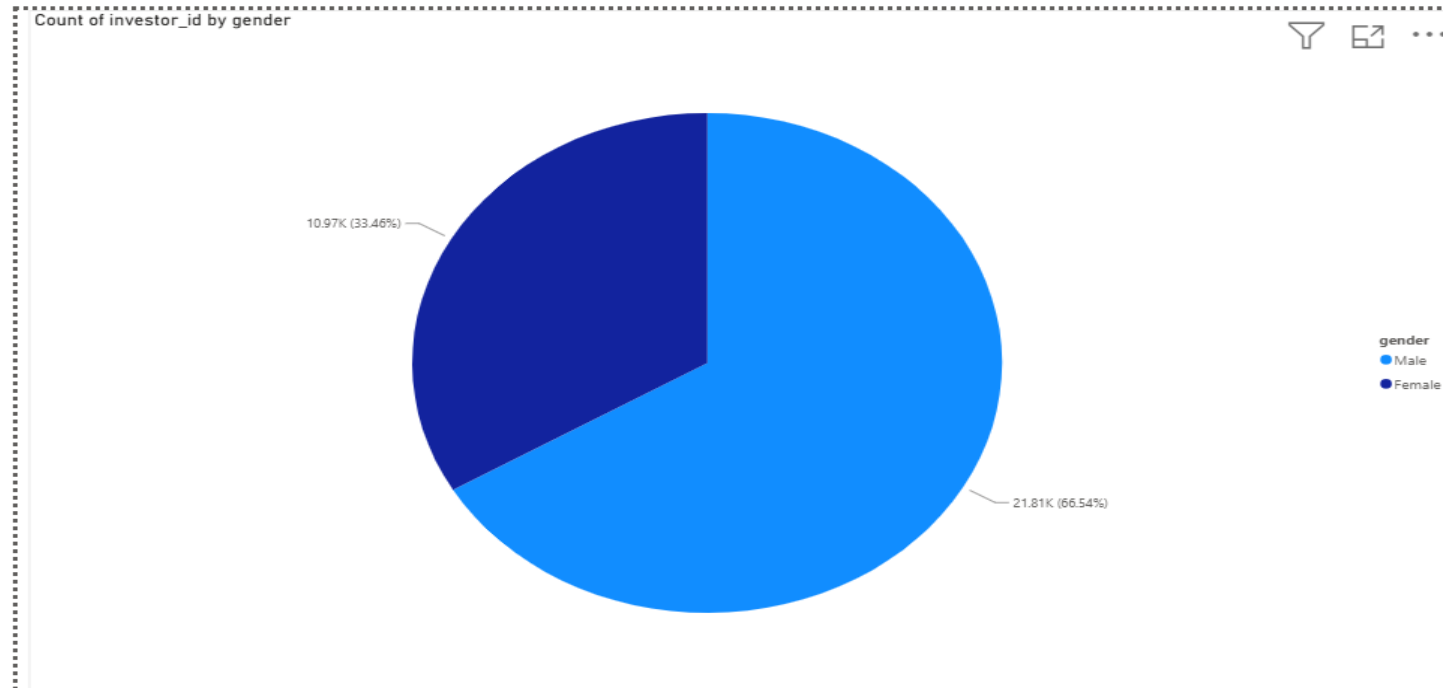
Key Insights:

- ▶ SIP inflows increased steadily from 2022 to 2025.
- ▶ Investor participation showed consistent growth.
- ▶ Mutual fund investments gained momentum over the analysis period.
- ▶ The trend reflects growing investor confidence in mutual funds.

AUM Growth by Fund House (2022-2025)



Investor Analytics Dashboard



Fund Category Distribution

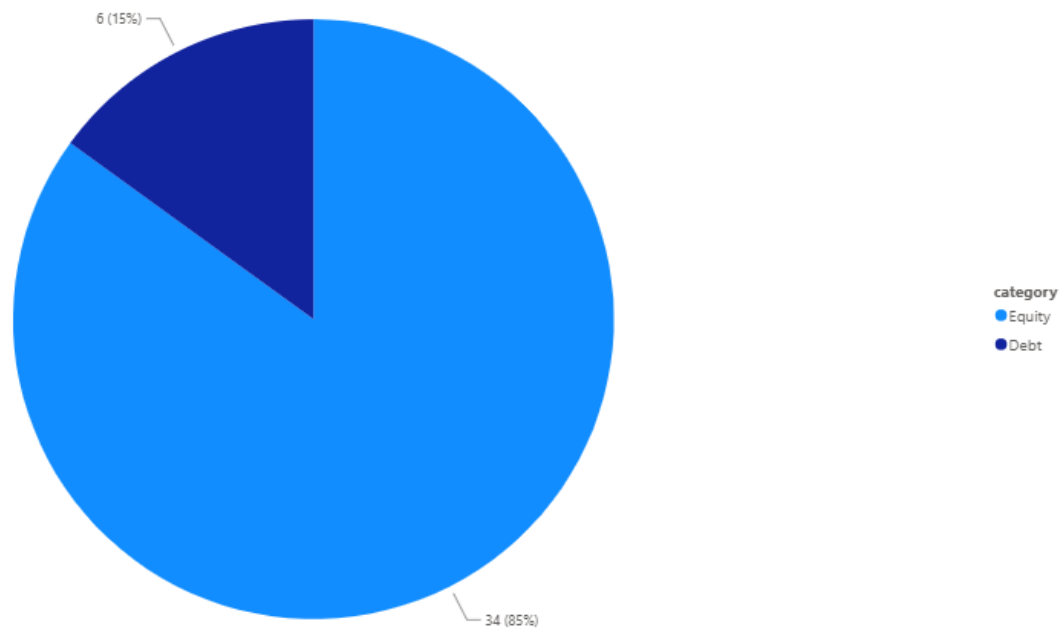


Fig.Fund Category Distribution

Key Insights:

Fund Category Distribution:

- ▶ Equity funds accounted for the majority of mutual fund schemes.
- ▶ Debt funds represented a smaller portion of the portfolio.

Investor Demographics:

- ▶ Male investors constituted approximately 66% of the investor base.
- ▶ Female investors accounted for around 34% of investors.
- ▶ The investor base showed healthy participation across demographics.

Business Value:

- ▶ Helps understand investor preferences.
- ▶ Supports fund segmentation and targeted investment strategies.
- ▶ Provides insights into market participation trends.

Risk & Performance Dashboard

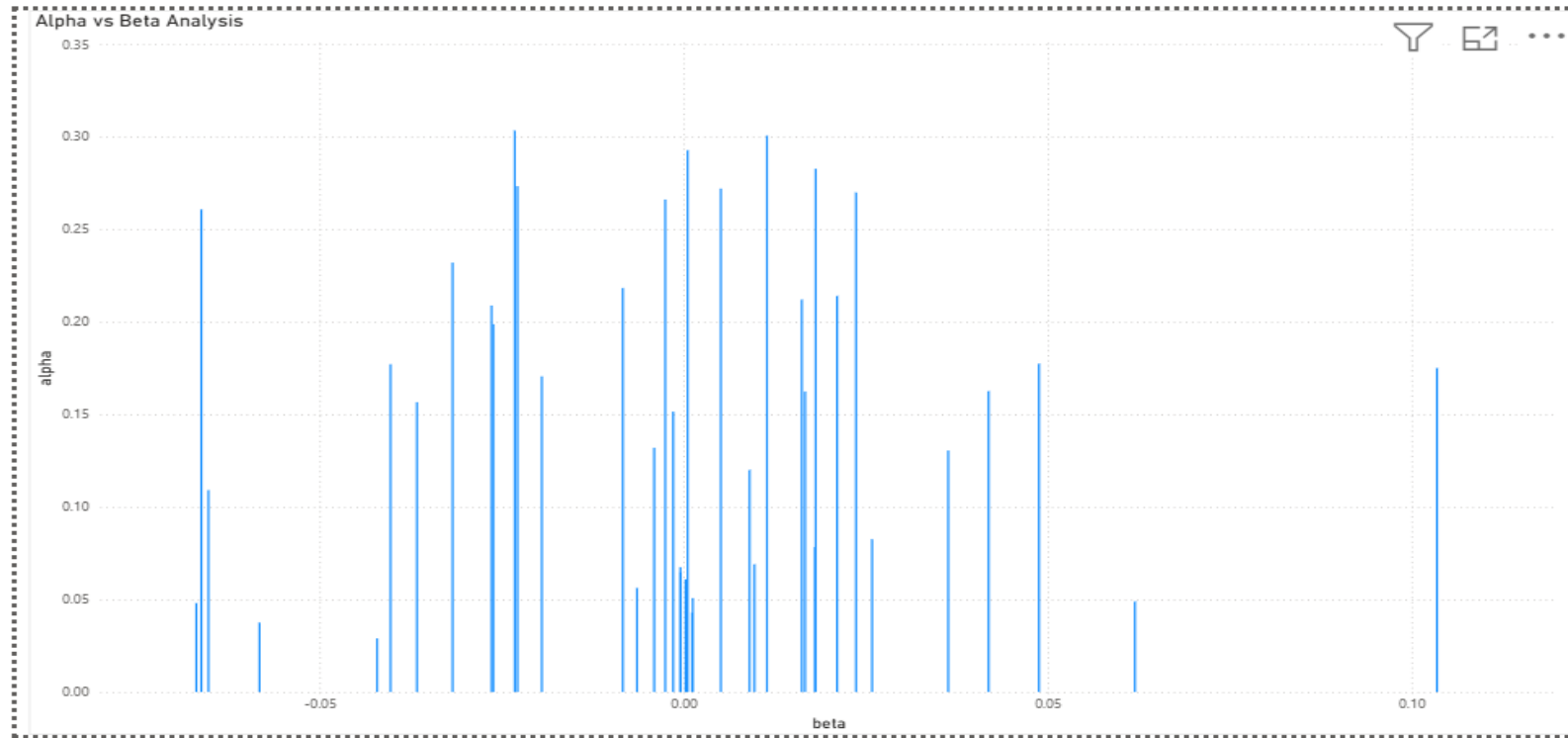


Fig.Alpha vs Beta Analysis

amfi_code	Sum of tracking_error
100033	0.23
119598	0.28
120505	0.23
120843	0.21
148567	0.19
Total	1.14

Fig.Tracking Error Analysis



Fig.Fund Score Distribution (Treemap)

Key Insights:

- ▶ Alpha and Beta metrics were used to evaluate benchmark-relative performance.
- ▶ Tracking Error measured benchmark tracking consistency.
- ▶ Fund Score Distribution highlighted top-performing mutual funds.
- ▶ Risk-adjusted analysis enabled comprehensive fund evaluation.
- ▶ The analytics framework enabled identification of funds with balanced risk-return characteristics.
- ▶ Comparative analysis supported more informed and data-driven investment decision-making.
- ▶ Performance metrics helped distinguish consistently performing funds from highly volatile schemes.
- ▶ Integrated risk analytics improved overall mutual fund evaluation accuracy.
- ▶ The dashboard simplified interpretation of complex financial performance indicators through interactive visualizations.

Key Project Insights

1. Growing SIP Investments:

- ▶ Monthly SIP inflows showed a consistent upward trend from 2022 to 2025.
- ▶ Indicates increasing investor confidence in mutual fund investments.

2. Equity Funds Dominate:

- ▶ Equity funds accounted for the majority of schemes in the dataset.
- ▶ Investors showed a stronger preference for long-term wealth creation through equity investments.

3. Strong Fund Performance:

- ▶ Several mutual funds demonstrated superior performance based on CAGR, Sharpe Ratio, and Sortino Ratio.
- ▶ Fund scorecard helped identify top-performing schemes.

4. Effective Risk Assessment:

- ▶ Alpha, Beta, Tracking Error, and Maximum Drawdown provided a comprehensive view of fund risk.
- ▶ Risk-adjusted metrics enabled better investment evaluation.

5. Dashboard-Driven Decision Making:

- ▶ Power BI dashboards transformed complex financial data into actionable insights.
- ▶ Improved fund comparison and performance monitoring.

Conclusion & Future Scope

Conclusion:

- ▶ Successfully analyzed mutual fund datasets using Python and Power BI.
- ▶ Performed data cleaning, exploratory analysis, performance evaluation, and risk assessment.
- ▶ Developed a fund scoring framework to rank mutual fund schemes.
- ▶ Built interactive dashboards for investor analytics and fund performance monitoring.
- ▶ Generated meaningful insights to support informed investment decisions.

Future Scope:

- ▶ Integrate real-time mutual fund and benchmark data APIs.
- ▶ Develop a personalized fund recommendation system.
- ▶ Implement predictive analytics using Machine Learning models.
- ▶ Add portfolio optimization and asset allocation features.

Thank You

Questions & Discussion

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